CHAPTER

58 Great vessels

MAJOR BLOOD VESSELS

The major blood vessels are the pulmonary trunk, the thoracic aorta and its branches, and the superior and inferior venae cavae and their tributaries.

ARTERIES

Pulmonary trunk

The pulmonary trunk, or pulmonary artery, conveys deoxygenated blood from the right ventricle to the lungs. About 5 cm in length and 3 cm in diameter, it is the most anterior of the cardiac vessels and arises from the pulmonary anulus surrounding the subpulmonary infundibulum at the base of the right ventricle, superior and to the left of the supraventricular crest. It slopes posterosuperiorly, at first anterior to the ascending aorta and then to its left; at the level of the fifth thoracic vertebra, inferior to the aortic arch at the left of the midline, it divides into right and left pulmonary arteries, of almost equal size. The bifurcation of the pulmonary trunk lies anteroinferior and to the left of the tracheal bifurcation and its associated inferior tracheobronchial lymph nodes and deep cardiac plexus. In the fetus, at the level of the bifurcation, the pulmonary artery is connected to the aortic arch by the ductus arteriosus.

Relations The pulmonary artery is entirely within the pericardium, enclosed with the ascending aorta in a common tube of visceral pericardium. The fibrous pericardium gradually disappears within the adventitia of the pulmonary arteries. Anteriorly, it is separated from the sternal end of the left second intercostal space by the pleura, left lung and pericardium. Posterior relations are the ascending aorta and left coronary artery initially, then the left atrium. The ascending aorta ultimately lies on its right. The appendage and coronary artery lie on each side of its origin. The superficial cardiac plexus lies between the pulmonary bifurcation and the aortic arch. The tracheal bifurcation, lymph nodes and nerves are superior, bilateral and to the right.

Variations and congenital conditions The pulmonary trunk is a relatively constant structure and there are minimal variations in healthy individuals. Congenital anomalies include pulmonary atresia and truncus arteriosus.

The coronary arteries usually originate from within the aortic sinuses, but occasionally may arise from ectopic locations. Most commonly, an ectopic left coronary artery arises from the pulmonary trunk or one of its branches (Bland–White–Garland syndrome). This potentially fatal condition requires urgent surgical correction because the myocardium is supplied with pulmonary blood instead of systemic blood (Loukas et al 2009). Infantile symptoms include pallor, fatigue, irritability, weak cry, cough, dyspnoea, and signs of ischaemia and cardiac failure precipitated by feeding, bowel movements or crying. The electrocardiogram may reveal deep narrow Q waves, left ventricular hypertrophy and left axis deviation. Radiologically, cardiomegaly is present with left ventricular and left atrial enlargement. Colour flow imaging can identify the anomalous origin of the left coronary artery. The pulmonary trunk may also give rise to the right coronary artery, the left interventricular (anterior descending) coronary artery, or even both coronary arteries. Typically, there is a large development of collateral vessels in the heart (**Figs 58.1**–**58.2**).

Pulmonary atresia is caused by a complete obstruction of pulmonary outflow and may be due to an absence or defect of the pulmonary valvular leaflets. It is associated with a blind-ending pulmonary trunk that causes right ventricular hypoplasia. Reduced flow may render the pulmonary trunk atretic, small or even normal in size, making diagnosis challenging. A diminished pulmonary flow is supplied through a patent ductus arteriosus (Smith and McKay 2004), and a concomitant ventricular septal defect may permit outflow from the right ventricle. Surgical repair is necessary to allow adequate oxygenation throughout the body.

In truncus arteriosus, a single arterial trunk exits the heart and subsequently divides into the pulmonary trunk and the aorta. Early neonatal life is possible because there is usually a coexisting ventricular septal defect; expedited surgical repair is necessary to avoid congestive heart failure, failure to thrive and death.

Right and left pulmonary arteries

The pulmonary arteries are described on page 959.

Thoracic aorta

Ascending aorta

The ascending aorta is typically 5 cm long. It originates at the base of the left ventricle, level with the inferior border of the third left costal cartilage, and ascends obliquely, curving anteriorly and to the right, and passing from posterior to the left half of the sternum to the level of the superior border of the second left costal cartilage. In children, the diameter of the thoracic aorta correlates most closely with body surface area (Kervancioglu et al 2006, Kaiser et al 2008).

Relations The ascending aorta lies within the fibrous pericardium, enclosed in a tube of serous pericardium together with the pulmonary trunk (see Figs 57.3, 57.4A–C). The infundibulum, initial segment of the pulmonary trunk and right appendage are anterior to its lower part. Superiorly, it is separated from the sternum by the pericardium, right pleura, anterior margin of the right lung, loose areolar tissue and the thymus or its remnants. The left atrium, right pulmonary artery and principal bronchus are posterior. The superior vena cava and right atrium, the former partly posterior, are to the right. The left atrium and, more superiorly, the pulmonary trunk are to the left. At least two aorticopulmonary bodies lie between the ascending aorta and the pulmonary trunk. The inferior aorticopulmonary body is near the heart and anterior to the aorta, and the middle aorticopulmonary body is near the right side of the ascending aorta.

The aortopulmonary window is a space between the pulmonary artery and aortic arch, bordered by the ascending aorta anteriorly, the descending aorta posteriorly, the mediastinal pleura laterally and the left principal bronchus medially (Deutsch and Savides 2005) (**Fig. 58.3**). It contains lymph nodes, fatty tissue, the ligamentum arteriosum and the left recurrent laryngeal nerve.

Aortic arch

The aortic arch continues from the ascending aorta (see Figs 57.3, 57.4A–C). Its origin, slightly to the right, is level with the superior border of the right second sternocostal joint. The arch first ascends diagonally and posteriorly to the left over the anterior surface of the trachea, then posteriorly across its left side, finally descending to the left of the fourth thoracic vertebral body, continuing as the descending thoracic aorta. It terminates level with the sternal end of the left second costal cartilage and so lies wholly within the superior mediastinum. It curves around the hilum of the left lung, and extends superiorly to the midlevel of the manubrium of the sternum.

The shadow of the arch is easily identified in frontal chest radiographs; its left profile is sometimes called the aortic 'knuckle' or 'knob' (see Fig. 56.16). There may also be an 'aortic nipple' (the shadow of the adjacent left superior intercostal vein crossing posteroanteriorly to its left). The aortic arch is best visualized in left anterior oblique views

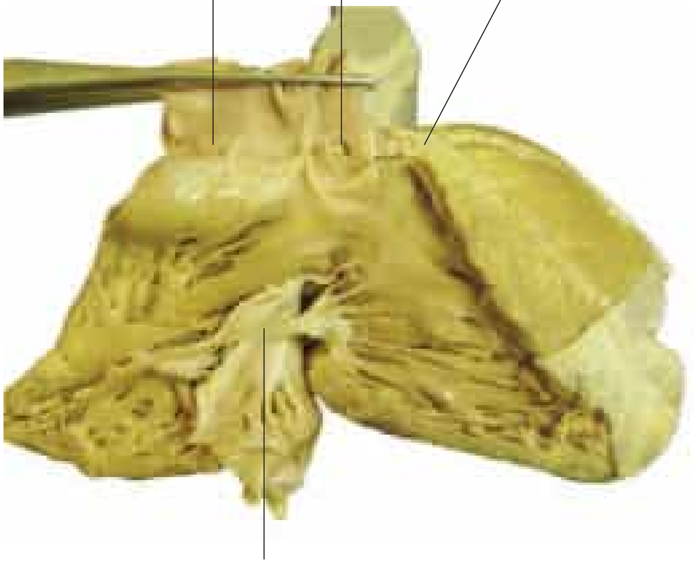

Fig. 58.1 A case of a left anterior interventricular artery arising from the pulmonary trunk. (Courtesy of Professor Loukas. With permission from Loukas M, Groat C, Khangura R, Owens DG, Anderson R. The normal and abnormal anatomy of the coronary arteries. Clinical Anatomy. 2009; 22: 114–128.)

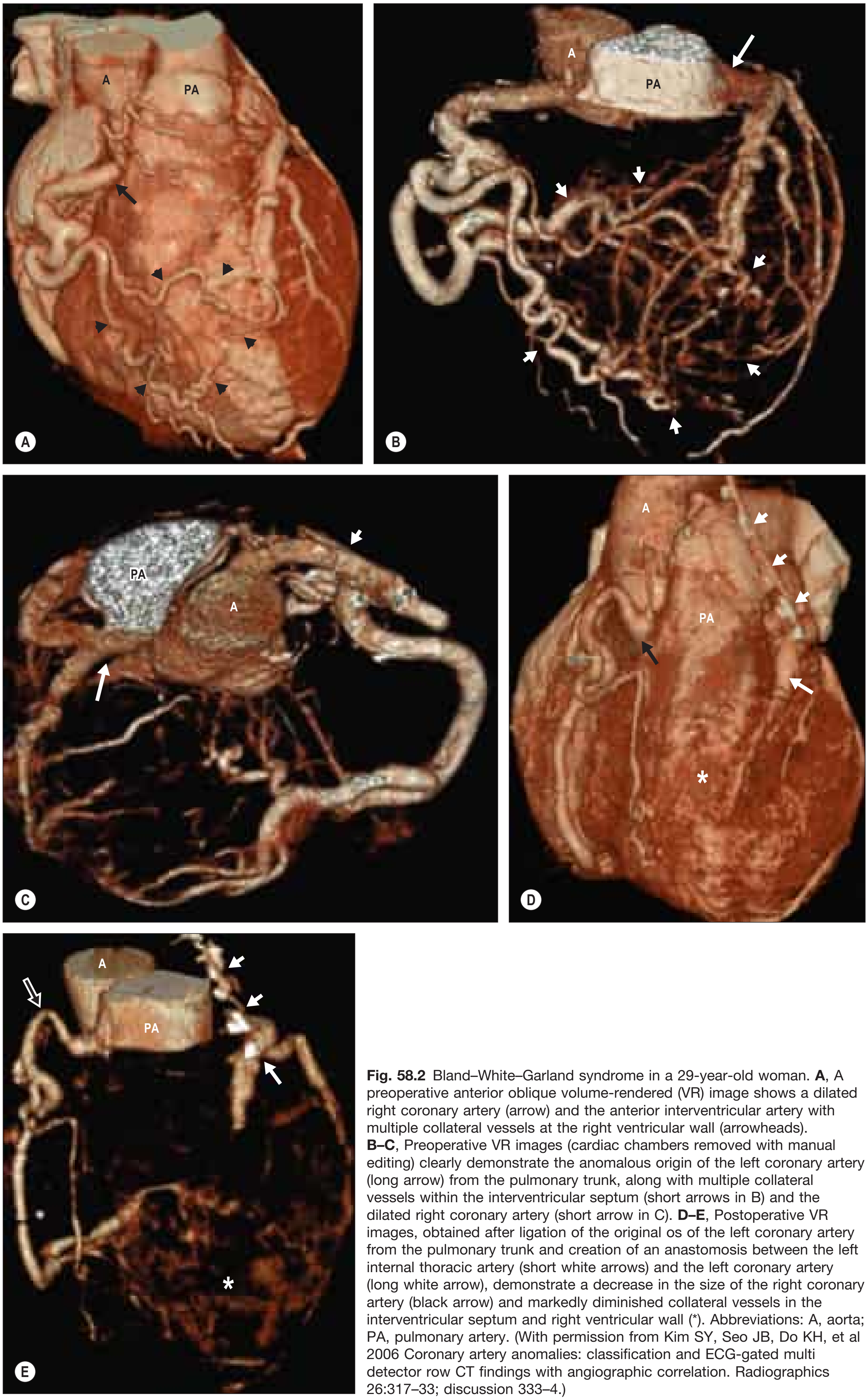

Fig. 58.2 Bland–White–Garland syndrome in a 29-year-old woman. **A**, A preoperative anterior oblique volume-rendered (VR) image shows a dilated right coronary artery (arrow) and the anterior interventricular artery with multiple collateral vessels at the right ventricular wall (arrowheads). **B–C**, Preoperative VR images (cardiac chambers removed with manual editing) clearly demonstrate the anomalous origin of the left coronary artery (long arrow) from the pulmonary trunk, along with multiple collateral vessels within the interventricular septum (short arrows in B) and the dilated right coronary artery (short arrow in C). **D–E**, Postoperative VR images, obtained after ligation of the original os of the left coronary artery from the pulmonary trunk and creation of an anastomosis between the left internal thoracic artery (short white arrows) and the left coronary artery (long white arrow), demonstrate a decrease in the size of the right coronary artery (black arrow) and markedly diminished collateral vessels in the interventricular septum and right ventricular wall (*). Abbreviations: A, aorta; PA, pulmonary artery. (With permission from Kim SY, Seo JB, Do KH, et al 2006 Coronary artery anomalies: classification and ECG-gated multi detector row CT findings with angiographic correlation. Radiographics 26:317–33; discussion 333–4.)

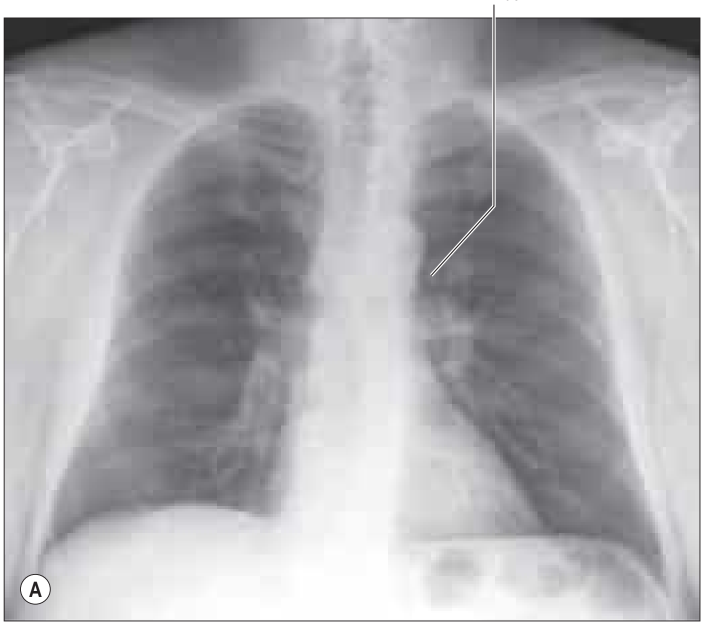

Fig. 58.3 The aortopulmonary (AP) window. **A**, A posteroanterior (PA) chest radiograph. **B**, A coronal computed tomography (CT) scan reconstruction.

on angiography and with equivalent computed tomography (CT) reconstruction planes; the pulmonary trunk and its left branch may be discerned nestling inferiorly in its concavity. The diameter of the arch initially matches that of the ascending aorta but is significantly reduced distal to the origin of the great vascular branches. The aortic isthmus, a small stricture at the border with the descending thoracic aorta, may be followed by a dilation; in the fetus, the isthmus lies between the origin of the left subclavian artery and the opening of the ductus arteriosus.

Relations The left mediastinal pleura is anterior and to the left of the arch. Deep to the pleura, the arch is crossed, in anteroposterior order, by the left phrenic nerve, left lower cervical cardiac branch of the vagus nerve, left superior cervical cardiac branch of the sympathetic trunk and the left vagus nerve (see Figs 57.3, 57.61). As the left vagus nerve crosses the arch, its recurrent laryngeal branch hooks below the vessel, to the left and behind the ligamentum arteriosum, then ascends on the right of the arch. The left superior intercostal vein ascends obliquely anteriorly on the arch, superficial to the left vagus nerve, deep to the left phrenic nerve. The left lung and pleura separate all these from the thoracic wall. Posterior and to the right are the trachea and the deep cardiac plexus, the left recurrent laryngeal nerve, oesophagus, thoracic duct and vertebral column. Superiorly, the brachiocephalic trunk, left common carotid and subclavian arteries arise from its convexity, and are crossed anteriorly near their origins by the left brachiocephalic vein. The pulmonary bifurcation, left principal bronchus, ligamentum arteriosum, superficial cardiac plexus and the left recurrent laryngeal nerve are all inferior. The concavity of the aortic arch, best viewed from the left, is the upper curved limit through which structures gain access to, or leave, the hilum of the left lung.

Pneumomediastinum and aortic nipple

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Variations of the arch The summit of the arch is usually about 2.5 cm inferior to the sternal notch but may diverge from this. In the infant, it is closer to the superior border of the sternum; the same is often the case in senescence as a result of vascular ectasia/unfolding.

In a right-sided aortic arch, the aorta curves over the right pulmonary hilum and descends to the right of the vertebral column; this is usually associated with transposition of the thoraco-abdominal viscera. Less often, after arching over the right hilum, the aorta may pass posterior to the oesophagus to gain its position (this is not accompanied by visceral transposition). The presence of a right-sided arch is of relevance to the paediatric surgeon who is planning repair of oesophageal atresia in neonates.

The aorta may divide into ascending and descending trunks, the former dividing into three branches to supply the head and upper limbs. It may divide near its origin, producing a double aortic arch; the two branches soon reunite and the oesophagus and trachea usually pass through the interval between them. If the aorta, ductus arteriosus or descending aorta entraps the oesophagus and/or trachea, this is referred to as a vascular ring.

Branches and variations Three branches arise from the convex aspect of the arch: the brachiocephalic trunk, left common carotid and left subclavian arteries (see Figs 57.3–57.4). They may branch from the beginning of the arch or the superior part of the ascending aorta. The distance between these origins varies, the most frequent being approximation of the left common carotid artery to the brachiocephalic trunk. Other branches may arise from the aortic arch, including the inferior thyroid, thyroidea ima, thymic, left coronary and bronchial arteries (Bergman et al 1988).

Coarctation of the aorta The aortic lumen is occasionally partly or completely obliterated, either above (preductal or infantile type), opposite or just beyond (postductal or adult type), the entry of the ductus arteriosus. In the preductal type, the length of coarctation is variable, aortic arch hypoplasia is common, and the left subclavian and even the brachiocephalic trunk may be involved. Severe forms of infantile coarctation and its extreme form (aortic interruption) may be patent ductus arteriosus-dependent, as there is no time for effective collateral circulation to develop. Prostaglandin infusion prior to transfer, and surgery at a tertiary centre often provide a very good mid- to long-term outlook for such infants.

The postductal type of coarctation has been attributed to abnormal extension of the ductal tissue into the aortic wall, stenosing both vessels as the duct contracts after birth. This form may permit years of normal life, allowing the development of an extensive collateral circulation to the aorta distal to the stenosis (**Figs 58.6–58.7**). High vascularity of the thoracic wall is important and clinically characteristic; many arteries arising indirectly from the aorta proximal to the coarctation segment anastomose with vessels connected with it distal to the block, and all of these vessels become greatly enlarged. Thus, in the anterior thoracic wall, the thoraco-acromial, lateral thoracic and subscapular arteries (from the axillary artery), the suprascapular artery (from the subclavian artery) and the first and second posterior intercostal arteries (from the costocervical trunk) anastomose with other posterior intercostal arteries, and the internal thoracic artery and its terminal branches anastomose with the lower posterior intercostal and inferior epigastric arteries.

Posterior intercostal arteries are always involved, and enlargement of their dorsal branches may eventually groove ('notch') the inferior margins of the ribs. The radiographic shadow of the enlarged left subclavian artery is also increased. Enlargement of the scapular vessels and anastomoses may lead to widespread interscapular pulsation (easily appreciated with the palm of the hand, and sometimes heard on auscultation).

Aortic aneurysm formation Degeneration of the aortic tunica media and intimal dissection play a major role in the pathogenesis of aneurysms affecting the ascending aorta and arch. Smooth muscle cells are lost and elastic fibres degenerate, producing cystic spaces in the media, which then fill with mucoid material. The loss of these structural cells leads to weakening of the wall with progressive dilation. Ageing and hypertension are major predisposing factors to cystic medial degeneration and there is a strong link with cigarette smoking. Thoracic aortic aneurysms are categorized by their location. A particular pattern of

Pneumomediastinum is an encompassing term that describes the presence of air in the mediastinum. It may arise from a wide range of pathological conditions or physiological states, e.g. penetrating trauma, ruptured major airways or oesophagus, hyperventilation or distressed ventilation such as acute asthma, periparturition or diabetic ketoacidosis. The 'aortic nipple' is the radiographic term used to describe a lateral nipple-like projection from the aortic knuckle seen in a small number of individuals that corresponds to the end-on appearance of the left superior intercostal vein coursing anteriorly. It may be mistaken radiologically for lymphadenopathy or an intrapulmonary nodule/neoplasm. Despite their relative independence, the aortic nipple is defined by new contours in cases of pneumomediastinum, taking on an 'inverted aortic nipple' appearance (Fig. 58.4). In this position, the inverted aortic nipple may facilitate radiographic discrimination of pneumomediastinum from similar conditions. An aortic nipple has been shown to precede pathological conditions such as venous obstruction in the superior and inferior vena cavae or left brachiocephalic vein, and has been identified in conditions where venous flow through the left superior intercostal vein is increased (e.g. portal hypertension and certain congenital venous anomalies).

There are several variants of this arrangement, of which the most common are as follows: a right-sided aortic arch and upper portion of the descending aorta pass anterior to the oesophagus and trachea, and the ductus arteriosus passes posterior to the oesophagus into an aortic diverticulum; a right-sided aortic arch and the upper portion of the descending aorta pass anteriorly, and the ductus arteriosus is inserted into the left subclavian artery, which arises as a fourth branch from the aortic arch; a left-sided descending aorta is attached to a left-sided ductus arteriosus anterior to the oesophagus, and a right-sided arch passes posteriorly; the right superior portion of the descending aorta wraps around the oesophagus and the ductus arteriosus is right-sided; or the right upper portion of the aorta wraps around the oesophagus and the ductus arteriosus is left-sided (Bergman et al 1988).

The right common carotid and subclavian arteries may arise separately, in which case the latter often branches from the left end of the arch distal to the left subclavian artery, and usually passes posterior to the oesophagus as an aberrant right subclavian artery. When this artery arises from a dilated portion of the descending aorta, the diverticulum of Kommerell, it is known as the lusoria artery (Fig. 58.5). It may become aneurysmal and cause fatal haemorrhage during endoscopy. It may form a vascular ring around the oesophagus and trachea, presenting in the neonate as a feeding disorder with failure to thrive.

The left vertebral artery may arise between the left common carotid and the subclavian arteries.

A rare avian form has been reported in which the right common carotid and subclavian arteries arise from the aortic arch, and the left common carotid and subclavian arteries arise from the descending aorta (Bergman et al 1988). Another rare avian form with two brachiocephalic trunks, the right trunk originating both common carotid arteries, and the left trunk originating both subclavian arteries, has been reported. Another rare order of the aortic branches is (in order from right to left): the right subclavian artery, left subclavian artery, followed by the right common carotid, and left common carotid arteries close together – almost forming a common carotid trunk.

The aortic arch may also curve behind the oesophagus and trachea, instead of in front, with variations in the aortic branches as well. Another rare avian form reported is with both carotid arteries originating from the same stem, and a left subclavian artery originating from the arch, while the right subclavian artery arises from the descending aorta. One or more of the ductus arteriosi may remain patent.

Very rarely, external and internal carotid arteries arise separately, the common carotid artery being absent on one or both sides, or both carotid arteries and one or both vertebral arteries may be separate branches. When a 'right aorta' occurs, the arrangement of its three branches is reversed and the common carotid arteries may have a single trunk. Other arteries may branch from it: most commonly, one or both bronchial arteries and the thyroidea ima artery.

Associations also exist between thoracic aortic aneurysm and other connective tissue diseases such as homocysteinuria and Ehlers–Danlos syndrome. A genetic relation is seen in familial thoracic aortic aneurysm syndrome (Baliga et al 2007). Rarely, aneurysms may occur as a result of Takayasu's arteritis, or infections within the aortic wall.

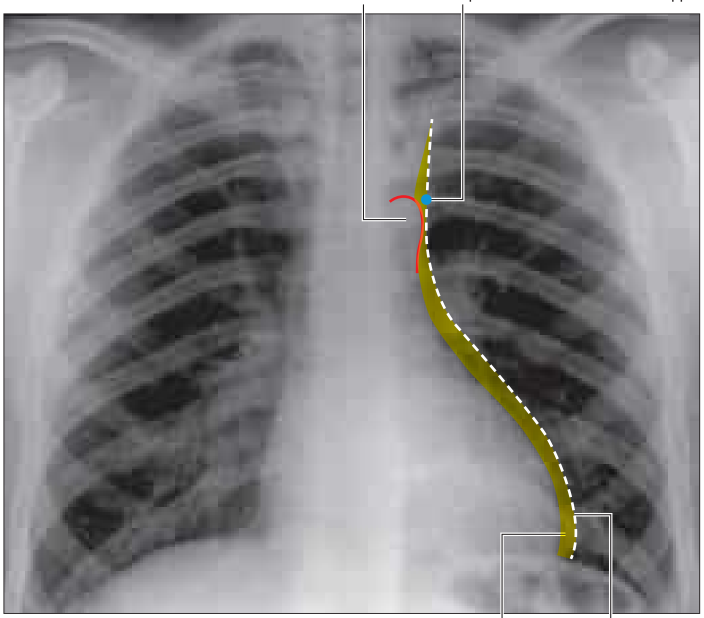

Fig. 58.4 A PA chest radiograph, anteroposterior view, showing the classic 'aortic nipple' in a patient with pneumomediastinum. A label is added on the left to delineate the nipple-like appearance of the left superior intercostal vein more clearly. (Courtesy of Professor Loukas. With permission from Walters A, Cassidy L, Muhleman M, Peterson A, Blaak C, Loukas M. Pneumomediastinum and the aortic nipple: the clinical relevance of the left superior intercostal vein. Clinical Anatomy 2013; 27:757–63.)

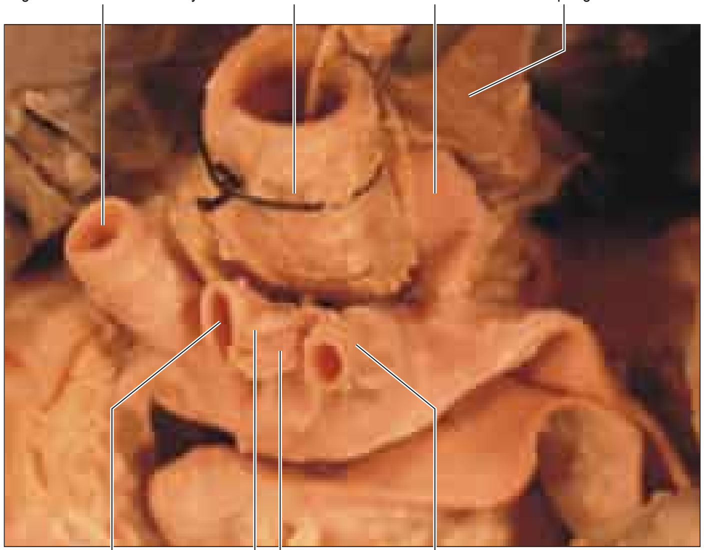

Fig. 58.5 A dissection showing a left aortic arch that gives rise to a common vertebral trunk and a retrotracheal right subclavian artery (RtRSA), also known as a lusoria artery. (Courtesy of Professor Loukas with permission from Loukas M, Louis RG Jr, Gaspard J, Fudalej M, Tubbs RS, Merbs W. A retrotracheal right subclavian artery in association with a vertebral artery and thyroidea ima. Folia Morphol (Warsz). 2006 Aug;65(3):236–41.)

CHAPTER 58

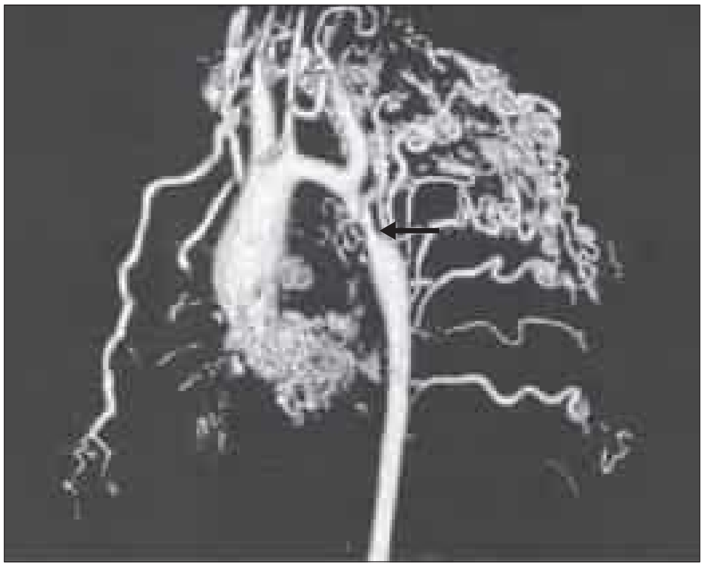

Fig. 58.6 A cardiac magnetic resonance (MR) angiograph showing three-dimensional reconstruction of a native aortic coarctation (arrow) in an adult with extensive collateral flow. Note the marked dilation of the left subclavian artery, supplying most of the collateral vessels, and mild aortic arch hypoplasia. (Courtesy of Dr Raad Mohiaddin, Royal Brompton Hospital, London.)

Fig. 58.7 The collateral circulation at MR angiography and phase-contrast velocity-encoded cine MRI. The MR angiogram shows the intercostal (short arrows) and internal thoracic (medium arrow) arteries, obtained in a patient with a haemodynamically significant aortic coarctation (long arrow). Note that the arteries appear enlarged. (With permission from Hom JJ, Ordovas K, Reddy GP. Velocity-encoded cine MR imaging in aortic coarctation: functional assessment of hemodynamic events. Radiographics 2008 Mar–Apr;28(2):407–16.)

aortic root involvement is seen in those with Marfan's syndrome, known as anulo-aortic ectasia (Baliga et al 2007). Descending aortic aneurysms are generally caused by atherosclerosis (90%); the remainder result from mycotic disease or trauma.

Some aortic aneurysms are incidental findings on chest films or CT studies. Symptomatic cases present with breathlessness, unbearable chest and back pain, hoarse voice, cough and haemoptysis. Early diastolic murmurs caused by aortic regurgitation may be audible. Repair is carried out in patients with symptoms or fusiform dilation measuring more than 5 cm in diameter.

Aneurysms may also occur in an aberrant right subclavian artery and may or may not involve the diverticulum of Kommerell, leading to dysphagia and even tracheal compression. Inadvertent rupture may occur during endoscopy because of its retro-oesophageal position.

Aortic dissection

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Aortopulmonary paraganglia

Aortopulmonary paraganglia (aortic bodies) belong to the branchiomeric category of paraganglia. They are chemoreceptors and respond to changes in arterial gas concentrations such as lowered pO_2, increased pCO_2 and increased hydrogen ion concentration. These microscopic structures occur in several locations within the thorax: coronary paraganglia lie between the ascending aorta and pulmonary trunk, either anteriorly or posteriorly, adjacent to the aortic root; pulmonary paraganglia lie in the groove between the ductus arteriosus and the pulmonary artery; and subclavian–supra-aortic paraganglia lie between either the right subclavian and right common carotid arteries or the left subclavian and left common carotid arteries, or caudal to the left subclavian artery, adjacent to the aortic arch.

Brachiocephalic trunk (artery)

The brachiocephalic (innominate) artery (trunk), the largest branch of the aortic arch, is 4–5 cm in length (see Fig. 57.4A,B). It arises from the convexity of the arch posterior to the centre of the manubrium of the sternum, and ascends posterolaterally to the right, at first anterior to the trachea, then on its right. The brachiocephalic and left common carotid arteries often share a common origin (Osborn 1998). It divides into the right common carotid and subclavian arteries level with the upper border of the right sternoclavicular joint.

Relations Sternohyoid and sternothyroid, thymic remnants, left brachiocephalic and right inferior thyroid veins, crossing its root, and sometimes the right cardiac branches of the vagus nerve, all separate the brachiocephalic trunk from the manubrium. Posterior are the trachea (superiorly) and the right pleura (inferiorly). The right vagus nerve is posterolateral before passing lateral to the trachea. On its right side are the right brachiocephalic vein and the upper part of the superior vena cava and pleura, and on its left side are thymic remains, the origin of the left common carotid artery, the inferior thyroid veins and the trachea.

Branches The brachiocephalic trunk usually has only terminal branches, the right common carotid and subclavian arteries. Occasionally, a thymic or bronchial branch, or a thyroidea ima artery arise from it. The thyroidea ima artery is a small and inconstant artery that may arise from the aorta, right common carotid, subclavian or internal thoracic arteries; it ascends on the trachea to the thyroid isthmus, where it terminates.

Descending thoracic aorta

The descending aorta is the segment of the thoracic aorta that is confined to the posterior mediastinum (see Figs 56.1–56.2, 56.6–56.7B, 56.13–56.14A, 56.19D,E). It begins level with the lower border of the fourth thoracic vertebra, continuous with the aortic arch, and ends anterior to the lower border of the twelfth thoracic vertebra in the aortic hiatus. At its origin, it is left of the vertebral column; as it descends, it reaches and terminates in the midline.

Relations Anterior to the descending thoracic aorta, from superior to inferior, are the left pulmonary hilum, the pericardium separating it from the left atrium, oesophagus and the vertebral portion of the diaphragm. The vertebral column and hemiazygos veins are posterior. Posterior to the right are the azygos vein and thoracic duct, and inferiorly, the right pleura and lung; the pleura and lung are to the left. The oesophagus with its plexus of nerves is located laterally and to the right in the upper thorax, anteriorly in the lower thorax and descends to a left anterolateral position when it reaches the diaphragm. To a limited degree, the descending aorta and oesophagus are mutually spiralized. Occasionally, the aorta enters the diaphragm to the left and behind the oesophagus (Berdjas and Turina 2011).

Surface anatomy The descending thoracic aorta may be projected as a 2.5 cm broad band from the sternal end of the second left costal cartilage to a median position 2 cm above the transpyloric plane at the first lumbar vertebra.

Branches

The thoracic aorta provides visceral branches to the pericardium, lungs, bronchi and oesophagus, and parietal branches to the thoracic wall.

Pericardial branches A few small vessels are distributed to the posterior aspect of the pericardium.

Bronchial arteries Bronchial arteries vary in number, size and origin. There is usually only one right bronchial artery; it arises from either the third posterior intercostal or the upper left bronchial artery, and runs posteriorly on the right principal bronchus. Its branches supply these structures, the pulmonary areolar tissue and the bronchopulmonary lymph nodes, pericardium and oesophagus. The left bronchial arteries, usually two, arise from the thoracic aorta – the upper near the fifth thoracic vertebra, the lower below the left principal bronchus – and run posterior to the left main bronchus; their branches are distributed as on the right.

Oesophageal branches Two or three bronchial arteries supply the thoracic portion of the oesophagus. In addition, two or three oesophageal arteries that arise either anteriorly or from the right side of the aorta supply the distal oesophagus (Norton et al 2008).

Mediastinal branches Numerous small vessels supply lymph nodes and areolar tissue in the posterior mediastinum.

Phrenic branches Superior phrenic arteries arise from the lower thoracic aorta and are distributed posteriorly to the superior diaphragmatic surface, anastomosing with the musculophrenic and pericardiacophrenic arteries.

Posterior intercostal arteries The posterior intercostal arteries and their branches are described on page 943.

Subcostal arteries Subcostal arteries are the last paired branches of the thoracic aorta, in series with the posterior intercostal arteries and inferior to the twelfth ribs. Each runs laterally anterior to the twelfth thoracic vertebral body and posterior to the splanchnic nerves, sympathetic trunk, pleura and diaphragm. The right subcostal artery is also posterior to the thoracic duct and azygos vein, while the left is posterior to the accessory hemiazygos vein. Each artery enters the abdomen at the lower border of the twelfth rib, accompanied by the twelfth thoracic (subcostal) nerve, lying posterior to the lateral arcuate ligament and kidney, and anterior to quadratus lumborum. The right artery courses posterior to the ascending colon, and the left courses posterior to the descending colon. Piercing the aponeurosis of transversus abdominis, each subcostal artery proceeds between this and internal oblique, and anastomoses with the superior epigastric, lower posterior intercostal and lumbar arteries. Each has a dorsal branch, distributed like those of the posterior intercostal arteries.

Aberrant artery A small artery sometimes leaves the descending thoracic aorta on its right near the right bronchial artery origin. It ascends to the right behind the trachea and oesophagus, and may anastomose with the right superior intercostal artery. It is a vestige of the right dorsal aorta and, occasionally, it is enlarged as the first part of a right subclavian artery.

Aortic rupture in trauma Rupture of the ascending aorta is usually associated with a high immediate mortality (Baliga et al 2007). Blunt aortic rupture commonly occurs in road traffic accidents and has a 20% survival rate. There is usually a transverse tear involving the tunica intima and tunica media of the aortic wall; systemic circulatory pressure

Aortic dissection occurs as a result of degeneration of the tunica media of the aortic wall and is associated with senescence, persistent hypertension or collagen vascular diseases such as Marfan's syndrome. Many patients with aortic dissection have a pre-existing aneurysm. Other associations include aortic coarctation, Turner's syndrome, cocaine abuse (<1%) and trauma during surgical procedures. There is a higher prevalence in men than women; after the age of 75, however, there is no sex difference (Baliga et al 2007). In a classic aortic dissection, an intimal tear may occur, producing a split into the tunica media that creates a false lumen (**Fig. 58.8**). If another tear occurs, connection can be made once again with the true lumen (the double-barrel aorta). Additional aetiopathologies include penetrating atherosclerotic ulceration, where an atherosclerotic plaque ruptures into the aortic tunica media, and aortic intramural haematoma, where the vasa vasorum haemorrhage into the wall of the aorta (Baliga et al 2007).

Aortic dissections are classified as types I, II, IIIa and IIIb (De Bakey classification), or types A and B (Stanford classification). Dissections that are proximal to the left subclavian artery, with or without distal extension, are classified as type I, II or A. Type I may involve the entire aorta, whereas type II involves the ascending aorta only. Dissections that are distal to the left subclavian artery are classified as type IIIa (involve the distal aorta up to the diaphragm), IIIb (extend below the diaphragm) or B. These cases present acutely with severe retrosternal, neck or interscapular chest pain. Depending on the extent of the dissection, they may be associated with neurological signs, diarrhoea or leg weakness. Extension into the pericardium causes cardiac tamponade and circulatory collapse. Diagnosis is established by echocardiography and on contrast-enhanced CT or magnetic resonance imaging (MRI). Medical management is possible for descending aortic dissections, while surgical repair is essential for ascending aortic or aortic arch dissection.

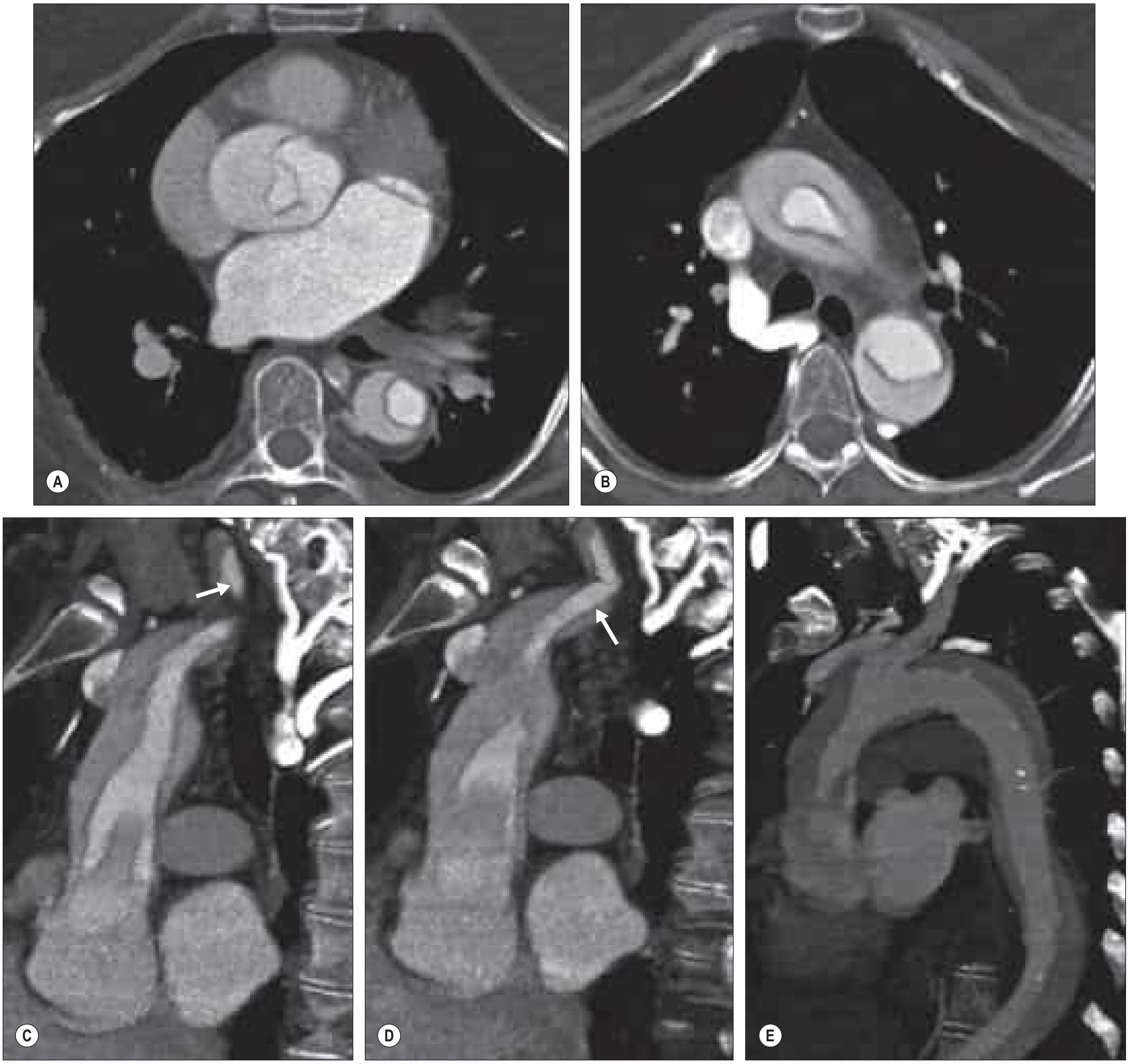

Fig. 58.8 A Stanford type A aortic dissection. **A–B**, Axial images taken through the thorax. **A**, Dissection flap involvement of the aortic root and descending aorta. **B**, The dissection flap in the aortic arch. **C–E**, A type A aortic dissection in a different patient. **C–D**, Sagittal reformatted images through the thorax and upper abdomen obtained at different levels from 64-slice multi-detector CT demonstrating intimal flap involvement of the ascending thoracic aorta. Note the extension of the dissection flap into aortic branch vessels (arrow). **E**, A sagittal maximum-intensity projection image showing dissection flap involvement of both the ascending and descending aorta. (With permission from McMahon MA, Squirrell CA. Multidetector CT of Aortic Dissection: A Pictorial Review. Radiographics. 2010 Mar;30(2):445–60.)

may cause the formation of a false aneurysm. Rupture of the isthmus region of the descending aorta is more common probably because it marks the junction between the mobile and fixed portions of the aorta. Other sites include the ascending aorta proximal to the origin of the brachiocephalic trunk, the aortic arch and the abdominal aorta. Rupture is likely to be the result of a number of factors, including torsion, shear and stretching forces, possibly compounded by hydrostatic pressure.

Aortic atherosclerosis or calcification Echocardiography, particularly trans-oesophageal, allows very detailed assessment of proximal aortic atherosclerosis implicated in systemic embolic events and strokes. The extent of turbulent flow may be documented. MRI may also allow an accurate assessment of the composition and size of atherosclerotic plaques and flow dynamics, permitting assessment of the risk of plaque rupture and thrombus formation.

Subclavian arteries

Right subclavian artery

The right subclavian artery arises from the brachiocephalic trunk, originating posterior to the upper border of the right sternoclavicular joint (see Fig. 51.2). It ascends superomedial to the clavicle and posterior to scalenus anterior, then descends laterally to the outer border of the first rib, where it becomes the axillary artery.

Left subclavian artery

In the majority of individuals, the left subclavian artery originates independently from the aortic arch after the origins of the brachiocephalic and left common carotid arteries (see Fig. 57.4). It rises into the neck lateral to the medial border of scalenus anterior, crosses posterior to this muscle and then descends towards the outer border of the first rib, where it becomes the left axillary artery. A common origin occasionally exists between the left subclavian and vertebral arteries. Rarely, there are bilateral brachiocephalic trunks, which subsequently divide on both sides into common carotid and subclavian arteries.

Relations In the thorax, the left subclavian artery is related anteriorly to the left common carotid artery and left brachiocephalic vein, from which it is separated by the left vagus, cardiac and phrenic nerves. More superficially, the anterior pulmonary margin, pleura, sternothyroid and sternohyoid lie between the vessel and the upper left area of the manubrium of the sternum. On the left side of the oesophagus, the thoracic duct and longus colli are posterior. The left subclavian artery is in contact posterolaterally with the left lung and pleura. The trachea, left recurrent laryngeal nerve, oesophagus and thoracic duct are medial. Laterally, the artery grooves the mediastinal surface of the left lung and pleura, which also encroach on its anterior and posterior aspects.

Common carotid arteries

The right and left common carotid arteries differ in their length and origin. The right is exclusively cervical and arises from the brachiocephalic trunk posterior to the right sternoclavicular joint. The left originates directly from the aortic arch immediately posterolateral to the brachiocephalic trunk and therefore has both thoracic and cervical parts.

Right common carotid artery

The right common carotid artery and its relations are described in Chapter 29.

Left common carotid artery

The left common carotid artery (see Figs 57.4, 29.7) ascends until level with the left sternoclavicular joint, where it enters the neck. Its thoracic portion is 20–25 mm long and it lies first anterior to the trachea, then inclines to the left. The further course of the artery is described in Chapter 29.

Relations Sternohyoid and sternothyroid, the anterior parts of the left pleura and lung, the left brachiocephalic vein and the thymic remnants are anterior and separate the left common carotid artery from the manubrium. The trachea, left subclavian artery, left border of the oesophagus, left recurrent laryngeal nerve and thoracic duct are posterior. To the right are the brachiocephalic trunk (inferior) and the trachea, inferior thyroid veins and thymic remains (superior). To the left are the left vagus and phrenic nerves, left pleura and lung.

VEINS

Brachiocephalic veins

The right and left brachiocephalic veins join to form the superior vena cava.

Right brachiocephalic vein

The right brachiocephalic vein is about 2.5 cm long. It arises posterior to the sternal end of the right clavicle and descends almost vertically to join the left brachiocephalic vein, forming the superior vena cava posterior to the inferior border of the first right costal cartilage, near the right sternal border. It is anterolateral to the brachiocephalic trunk and right vagus nerve; the right pleura, phrenic nerve and internal thoracic artery are initially posterosuperior, becoming lateral inferiorly (see Fig. 29.14). Its tributaries are the right vertebral, internal thoracic and inferior thyroid veins, and sometimes the first right posterior intercostal veins.

Left brachiocephalic vein

The left brachiocephalic vein is about 6 cm long, over twice the length of the right. It arises posterior to the sternal end of the left clavicle, anterior to the cervical pleura, and descends obliquely to the right, posterior to the superior half of the manubrium sterni, reaching the sternal end of the first right costal cartilage where it joins the right brachiocephalic vein to form the superior vena cava (see Fig. 29.14). It is separated from the left sternoclavicular joint and manubrium by sternohyoid and sternothyroid, the thymus or its remnants, and areolar tissue; terminally, it is overlapped by the right pleura. It crosses anterior to the left internal thoracic, subclavian, brachiocephalic and common carotid arteries, left phrenic and vagus nerves, and the trachea. The aortic arch is inferior to it. Its tributaries are the left vertebral, internal thoracic, inferior thyroid and superior intercostal veins, and sometimes the first left posterior intercostal, thymic and pericardial veins.

In children, the length and diameter of the left brachiocephalic vein are closely related to height; the diameter reaches adult dimensions at the age of 10 years (**Figs 58.9–58.10**) (Sanjeev and Karpawich 2006).

Superior vena cava

The superior vena cava returns blood to the heart from the tissues above the diaphragm. It is approximately 7 cm in length, and is formed by the junction of the brachiocephalic veins posterior to the lower border of the first right costal cartilage. It descends vertically, posterior to the first and second intercostal spaces, and drains into the upper right atrium posterior to the third right costal cartilage. Its inferior half is within the fibrous pericardium, which it pierces level with the second costal cartilage. Covered anterolaterally by serous pericardium (from which a retrocaval recess projects), it is slightly convex to the right (see Figs 57.3, 57.4A,B, 57.12, 29.14). The superior vena cava is valveless. In children, the length and diameter of the superior vena cava are also closely related to height; the diameter reaches adult dimensions at the age of 10 years (see Figs 58.9–58.10) (Sanjeev and Karpawich 2006).

Relations The anterior margins of the right lung and pleura are anterior and the pericardium intervenes below; these structures separate the superior vena cava from the right internal thoracic artery, first and second intercostal spaces, and second and third costal cartilages. The trachea and right vagus nerve are posteromedial, the right lung and pleura are posterolateral, and the right pulmonary hilum is posterior. The right phrenic nerve and pleura are immediate right lateral relations and the brachiocephalic trunk and ascending aorta lie to the left, the aorta overlapping the superior vena cava (see Fig. 55.7).

Tributaries Tributaries of the superior vena cava are the azygos vein and small veins from the pericardium and other mediastinal structures (see Fig. 56.7A).

Superior vena cava obstruction Superior vena cava obstruction is characterized by headaches, facial and neck venous congestion, and oedema, reflecting impaired venous drainage of the head, neck and arms, and of the collateral circulation, resulting in chest wall telangiectasia. Several of the symptoms may subside with recumbency, or may be aggravated by standing up. The obstruction may be either partial or complete, and may occur suddenly or gradually. It is usually caused by mediastinally invasive right upper lobe primary bronchogenic carcinoma or by metastatic involvement of the right paratracheal lymph nodes.

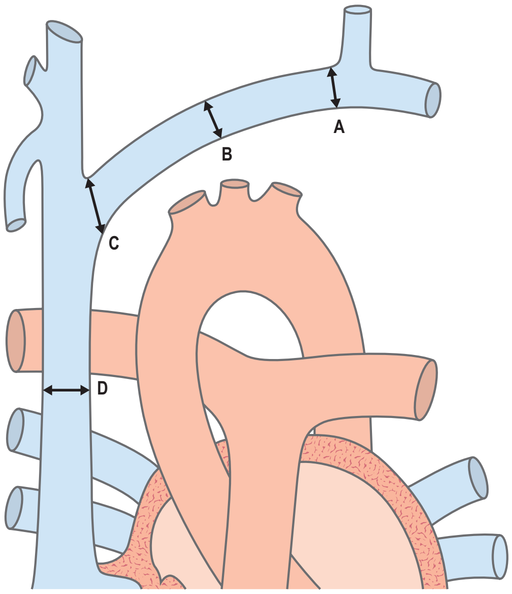

Fig. 58.9 Sites of measurement of the left brachiocephalic (innominate) vein and superior vena cava at angiography. Key: A, distal brachiocephalic vein; B, mid-brachiocephalic vein; C, brachiocephalic–superior vena cava junction; D, mid-superior vena cava. (Redrawn with permission from Sanjeev S, Karpawich PP. Superior vena cava and innominate vein dimensions in growing children: an aid for interventional devices and transvenous leads. Pediatr Cardiol. 2006 Jul–Aug;27(4):414–9.)

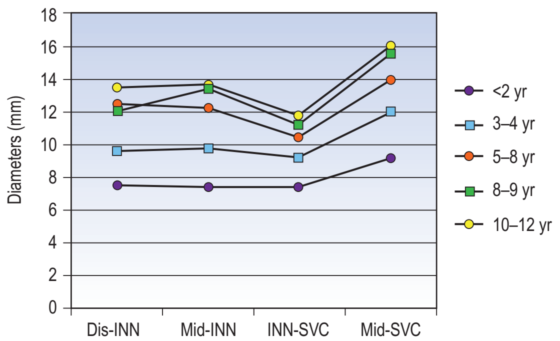

Fig. 58.10 Diameters of the distal brachiocephalic (innominate) vein (dis-INN), mid-brachiocephalic vein (mid-INN), brachiocephalic–superior vena cava junction (INN-SVC) and mid-superior vena cava (mid-SVC) in children of varying ages. (Redrawn with permission from Sanjeev S, Karpawich PP. Superior vena cava and innominate vein dimensions in growing children: an aid for interventional devices and transvenous leads. Pediatr Cardiol. 2006 Jul–Aug;27(4):414–9.)

Haemorrhagic intrathoracic goitre, mediastinitis (either benign, malignant or fibrous), mediastinal haematoma, constrictive pericarditis, mediastinal cyst, thrombosis of the superior vena cava and sarcoidosis are less common aetiologies. Radiographic findings might include widening of the upper mediastinum on the frontal film or obliteration of the retrosternal space on the lateral film. Other correlative findings may include pulmonary or mediastinal masses, lymphadenopathy, enlarged or obliterated azygos venous system, pleural effusion or rib notching. This is usually considered to be an oncological emergency and symptoms are often completely and promptly relieved by insertion of a vascular stent via the common femoral vein or by radiotherapy to the affected region after a tissue diagnosis is established.

Variations of the brachiocephalic veins and superior vena cava

Available with the *Gray's Anatomy* e-book

Inferior vena cava

The inferior vena cava returns blood to the heart from infradiaphragmatic tissues. It passes through the diaphragm between the right leaf and central area of the central tendon of the diaphragm at the level of the eighth and ninth thoracic vertebrae, and drains into the inferoposterior part of the right atrium (see Fig. 55.1, 57.4B). The thoracic part is very short, and is partly inside and partly outside the pericardial sac. The extrapericardial part is separated from the right pleura and lung by the right phrenic nerve, and the intrapericardial part is covered, except posteriorly, by inflected serous pericardium. The abdominal course of the inferior vena cava is described in Chapter 62.

Collateral venous channels In obstruction of the upper inferior vena cava, the azygos and hemiazygos veins and vertebral venous plexuses are the main collateral channels that maintain venous circulation. They connect the superior and inferior venae cavae and communicate with the common iliac vein by the ascending lumbar veins and with many tributaries of the inferior vena cava.

Variations Broadly speaking, the inferior vena cava may be congenitally interrupted, duplicated, left-sided or even absent.

Bonus e-book images

Fig. 58.1 A case of a left anterior interventricular artery arising from the pulmonary trunk.

Fig. 58.2 Bland–White–Garland syndrome in a 29-year-old woman.

Fig. 58.4 A PA chest radiograph, anteroposterior view, showing the classic 'aortic nipple' in a patient with pneumomediastinum.

Fig. 58.5 A dissection showing a left aortic arch that gives rise to a common vertebral trunk and a retrotracheal right subclavian artery, also known as a lusoria artery.

Fig. 58.6 A cardiac magnetic resonance angiograph showing three-dimensional reconstruction of a native aortic coarctation in an adult with extensive collateral flow.

Fig. 58.7 The collateral circulation at MR angiography and phase-contrast velocity-encoded cine MR imaging.

Fig. 58.8 A Stanford type A aortic dissection.

Fig. 58.9 Sites of measurement of the left brachiocephalic (innominate) vein and superior vena cava at angiography.

Fig. 58.10 Diameters of the distal brachiocephalic (innominate) vein mid-brachiocephalic vein, brachiocephalic–superior vena cava junction and mid-superior vena cava in children of varying ages.

The brachiocephalic veins may enter the right atrium separately, the right vein descending like a normal superior vena cava. During embryological development, if the cardinal vein does not become obliterated, it will persist as a left-sided superior vena cava. A left superior vena cava may have a slender connection with the right and then cross the left side of the aortic arch to pass anterior to the left pulmonary hilum before turning to enter the right atrium. It replaces the oblique vein of the left atrium and coronary sinus, and receives all the tributaries of the coronary sinus. The left brachiocephalic vein sometimes projects above the manubrium (more frequently in childhood), and crosses the suprasternal fossa in front of the trachea. A left-sided superior vena cava may cause difficulties when placing a cardiac catheter, pacing or defibrillating electrodes because the angle between the left superior vena cava and the left subclavian vein is much more acute than that between the subclavian and a normal left brachiocephalic vein. Its calibre may also be smaller than that of the right. Even if insertion of a catheter into a left superior vena cava is possible, the angle at which the catheter enters the right atrium causes difficulty when attempting to place it into the right ventricle and pulmonary trunk; this generally leaves the catheter tip against the coronary sinus, making it difficult to obtain blood samples. In the majority of persistent left superior venae cavae, blood drains into the right atrium via the coronary sinus. When this does not happen, the coronary sinus is absent, and the persistent left superior vena cava drains directly into the atrium. Cyanosis reflects a persistent right-to-left shunt and affected individuals have a higher risk for paradoxical embolism. Radiographically, there is a paramediastinal bulge before the aortic arch. Electrocardiograms show a left P-wave axis. Diagnosis can be confirmed by cardiac catheterization and angiography. Persistent left superior vena cava may be associated with extracardiac conditions such as VACTERL association (vertebral defects, anal atresia, cardiac malformations, tracheo-oesophageal fistula with oesophageal atresia, radial and renal dysplasia, and limb anomalies); trisomy 21; 22q11; CHARGE association (coloboma, heart defects, choanal atresia, mental retardation, genital and ear anomalies); and 45 XO karyotype (Turner's syndrome).

If the superior vena cava is duplicated, the right superior vena cava may drain into the right atrium and the left superior vena cava into the left atrium. Although double superior vena cava is largely asymptomatic, it is usually (along with a sole left-sided superior vena cava) associated with congenital cardiac anomalies, such as pulmonary stenosis, coarctation of the aorta, tetralogy of Fallot and atrial septal defects.

Embryologically, if the right subcardinal vein fails to anastomose with the hepatic sinusoids, the hepatic segment of the inferior vena cava fails to develop. When this occurs, the hemiazygos or azygos veins, which both originate from the cranial supracardinal veins, return blood to the heart. Azygos continuation of the inferior vena cava, characterized by a prominent azygos vein, is an entity of which the paediatric surgeon should be aware when undertaking repair of oesophageal atresia and tracheo-oesophageal atresia in neonates. The azygos vein, which is commonly ligated and divided during this procedure, should be spared because ligation of this vessel in neonates with azygos continuation of the inferior vena cava leads to intraoperative circulatory collapse and death. Double inferior vena cava (right side usually dominant) is a result of the persistence of all or any segments of the subcardinal veins. One of the most common variations is a left-sided inferior vena cava, which is formed if the right supracardinal vein regresses and the left supracardinal vein persists. There may also be a duplication of the inferior vena cava, which typically occurs below the renal veins, and is the result of persistence of the left lumbar and thoracic supracardinal veins and left suprasubcardinal anastomosis as well as malformation of right subcardinal–hepatic anastomosis. Abnormalities associated with duplication of the inferior vena cava include: congenital heart disease, congenital absence of the right kidney, cloacal extrophy, renal ectopia with abdominal aneurysm, right retrocaval ureter, left retrocaval ureter and congenital absence of the iliac anastomosis, abnormal left arm drainage and transcaval ureter.

In situs inversus with dextrocardia, the inferior vena cava passes inferiorly along the left side instead of the right (as opposed to situs inversus with levocardia, in which the inferior vena cava remains on the right side). In situs ambiguus, otherwise known as heterotaxy, the major organs are arranged ambiguously within the body. Prominent azygos veins with interrupted inferior vena cava are associated with heterotaxy (Punn and Olson 2010). Situs ambiguus with polysplenia is associated with an absence of portions of the inferior vena cava on the right side (with continuation through the azygos or hemiazygos vein), transposition of the inferior vena cava to the left side, or duplication of the inferior vena cava with one on the right and one on the left. Situs inversus with asplenia is similar to that of polysplenia, in that the inferior vena cava may be absent with azygos continuation or with the inferior vena cava to the left of the midline. In situs inversus totalis, transposition of the inferior vena cava consists of a right-sided inferior vena cava. The inferior vena cava may have an abnormally high insertion into the right atrium. Congenital agenesis of the inferior vena cava may occur and may be completely asymptomatic because venous drainage of the lower limbs occurs through anastomosed channels of the azygos and hemiazygos veins. That said, this condition may be associated with a higher risk for deep vein thrombosis.

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